

PRISM-SIF: Comprehensive Implementation Manual for Serious Injury and Fatality Precursor Detection in the Oil and Gas Sector

1. Academic Literature and Contextual Baseline for Code-Switched NLP

The architectural foundation of the PRISM-SIF (Predictive Recognition of Incidents and Safety Management for Serious Injury and Fatality) system requires a rigorous synthesis of applied natural language processing (NLP) in high-risk industrial domains and state-of-the-art research in code-switched computational linguistics. Unstructured textual narratives—spanning formal accident investigations, near-miss self-reports, shift handover logs, and permit-to-work documentation—contain the critical sequence of events representing the causal factors of industrial incidents¹. Historically, the extraction of actionable hazard intelligence from these reports relied entirely on manual analysis by safety engineers or rudimentary, rules-based keyword matching. However, recent advancements in deep learning architectures, specifically the deployment of transformer-based models for Named Entity Recognition (NER) and sequence classification, have demonstrated exceptional capability in parsing aviation and industrial safety corpora³. Systems such as Aviation-BERT-NER have achieved precision scores exceeding 95% by utilizing template-based fine-tuning over domain-specific structured datasets, demonstrating the viability of domain-adapted transformers for safety taxonomy extraction⁴.

The primary operational challenge in deploying a comparable system for Oil India Limited (OIL) lies in the linguistic realities of field data. Operational reports in the Indian subcontinent are rarely recorded in standardized, monolingual English. Instead, they are heavily code-switched, predominantly utilizing a fluid mixture of English, Hindi, and Assamese, frequently transcribed in Roman script via phonetic typing. Code-switching presents a severe mathematical and linguistic challenge for traditional language models due to structural ambiguity, lexical borrowing, non-standard orthography, and intra-word code-mixing⁵. When native languages mix dynamically with English technical jargon, models relying on monolingual pre-training or rigid subword tokenizers suffer catastrophic performance degradation.

To surmount this barrier, cross-lingual masked language models such as XLM-RoBERTa (XLM-R) have emerged as the academic and industry standard. Unlike earlier multilingual models such as mBERT or the original XLM, XLM-R explicitly discards the use of language embeddings during its unsupervised pre-training phase⁶. This architectural decision forces the model to rely

strictly on cross-lingual subword alignments and contextual representations, which paradoxically enhances its robustness to heavily code-switched text where explicit, rigid language boundaries do not exist⁶. XLM-R was pre-trained on 2.5 terabytes of filtered CommonCrawl data spanning 100 languages, utilizing a massive SentencePiece vocabulary of 250,000 tokens⁷. Empirical studies demonstrate that XLM-R outperforms mBERT on cross-lingual classification by up to 23% in accuracy for low-resource languages, yielding a 2.4% average F1-score improvement on named entity recognition tasks globally⁶. Recent benchmarking efforts by AI4Bharat have deeply contextualized the performance of these cross-lingual models on Indic languages. The Indian NLP landscape now encompasses over 200 datasets and 50 benchmarks addressing core linguistic processing tasks⁸. Studies on tokenization, normalization, and morphological analysis indicate that standard subword tokenization often improperly fragments Indic words due to their rich, agglutinative morphology and complex sandhi rules⁸. Research into Assamese and Nagamese NLP, including the development of datasets for zero-shot transfer and part-of-speech (POS) tagging, underscores the necessity of continuous pre-training or parameter-efficient fine-tuning on regional linguistic corpora⁸. Experiments on code-mixed datasets, such as Malayalam-English and Hinglish corpora, consistently prove that while standard models fail at zero-shot transfer on phonetic vernacular, fine-tuned XLM-R architectures set new state-of-the-art benchmarks in handling mixed linguistic origins⁵. The PRISM-SIF system fundamentally operates on the premise that XLM-R, augmented with localized Assamese-Hindi-English safety vocabulary and fine-tuned via weak supervision, serves as a highly accurate Layer 1 extractor for SIF precursors in the OIL operational context.

2. Data Acquisition Strategy and Regulatory Sources

The efficacy of a neuro-symbolic AI system is directly proportional to the quality, domain-specificity, and volume of its training data. PRISM-SIF necessitates a composite dataset that bridges highly structured international catastrophic incident data with localized, region-specific operational reports governing the Indian hydrocarbon sector.

International Process Safety Data Integration

To establish the foundational semantic understanding of process safety concepts, major hazard facilities, and catastrophic failure modes, PRISM-SIF ingests high-density unstructured data from premier international safety regulators. The U.S. Chemical Safety and Hazard Investigation Board (CSB) provides exhaustive final investigation reports, public meeting transcripts, and detailed root cause analysis documents regarding catastrophic chemical fires, toxic gas releases, and combustible dust explosions¹¹. These unstructured PDFs are critical for training the large language model (LLM) residing in Layer 3 on the complex, multi-stage causal chains of process safety failures.

Similarly, the Occupational Safety and Health Administration (OSHA) accident investigation summaries offer a high-volume corpus of brief narrative texts detailing severe, acute workplace injuries, providing vital data for training the sequence classification head of Layer 1. For

offshore-specific hazards, such as well-integrity failure modes, blow-out preventer (BOP) testing anomalies, and subsea isolation valve failures, the Bureau of Safety and Environmental Enforcement (BSEE) incident databases (including deep-water failure investigations like the Macondo blowout) serve as crucial training nodes¹². To ground the probabilistic weighting of the Causal Knowledge Graph (Layer 2), the system ingests data from the EPA Risk Management Plan (RMP) database, which mandates high-hazard facilities to submit historical accident data and off-site consequence analyses¹³. An aggregation of 15 major process safety incidents spanning from 1984 to 2022, accounting for over 4,000 fatalities, serves as the benchmark dataset for validating the system's ability to map catastrophic failure chains¹².

Indian Statutory Frameworks and OISD Integration

To adapt the model to the operational, legal, and compliance realities of the Indian oil and gas sector, regional statutory frameworks must be deterministically encoded into the system. The Oil Industry Safety Directorate (OISD) standards dictate the prescriptive safety requirements for the Indian hydrocarbon sector, acting as the ground-truth rulebook for the neuro-symbolic graph.

Key documents ingrained into the system include OISD-STD-105, which constitutes the definitive guide for the Work Permit System across hydrocarbon processing and handling installations¹⁶. This standard outlines the exact authorization protocols, required isolations, and explanatory notes for issuing cold work, hot work, and confined space entry permits¹⁶.

Furthermore, OISD-STD-117 establishes the foundational guidelines for fire protection facilities at petroleum depots, terminals, and pipeline installations¹⁷. It mandates the design criteria for fire water systems, automatic actuated foam flooding for floating roof tanks, halon protection systems, and minimum layout separation distances¹⁷. Directives from the Directorate General of Mines Safety (DGMS) and the Petroleum and Explosives Safety Organisation (PESO) regarding explosives handling and drilling safety are also integrated to define strict compliance boundaries.

To capture the localized code-switched vernacular, near-miss reporting styles, and historical incident data specific to Indian Public Sector Undertakings (PSUs), Right to Information (RTI) Act requests must be systematically filed and processed. These requests specifically target sanitized accident investigation board reports, root cause analysis (RCA) findings, and hazard identification (HAZID) logs from comparable domestic operators. This RTI-acquired data provides the vital, linguistically representative code-switched training inputs required to calibrate the XLM-R embeddings to the actual phonetic typing styles utilized by frontline workers in Assam and adjacent operational sectors.

3. Technology Stack and Infrastructure Architecture

The PRISM-SIF architecture is deployed as a sophisticated 3-layer neuro-symbolic pipeline, deliberately engineered for on-premise execution to guarantee data sovereignty, high-throughput batch processing, and privacy compliance. To minimize infrastructure capital expenditures while maximizing computational efficiency, the system is designed to operate

concurrently on a single NVIDIA A100 (80GB VRAM) instance, utilizing strict Docker Compose orchestration and memory partitioning.

Layer 1: Weak Supervision and XLM-RoBERTa

Layer 1 is responsible for the ingestion of raw, noisy text and the simultaneous execution of NER and sequence classification. Generating tens of thousands of manually annotated, code-switched training examples for deep learning is cost-prohibitive and glacially slow. Therefore, PRISM-SIF relies on a weak supervision framework to mathematically aggregate noisy, heuristic labeling functions into probabilistic training labels.

An architectural evaluation of weak supervision frameworks reveals significant operational differences:

- **Snorkel:** The historical industry standard for data programming. Snorkel utilizes a generative matrix factorization approach to denoise and aggregate labeling functions by learning their latent accuracies. While highly accurate, its generative model can be computationally heavy and slow during the iterative development of localized dictionaries.
- **FlyingSquid:** This framework relies on a triplet method for learning the accuracies of labeling functions without requiring a generative model. It offers execution speeds magnitudes faster than Snorkel, making it ideal for rapid prototyping, though it assumes statistical independence among labeling functions.
- **SPEAR (AI4Bharat):** An advanced toolkit optimized for subset selection, data programming, and semi-supervised learning, specifically tailored by researchers for Indic language challenges.

For the PRISM-SIF implementation, SPEAR is selected. Its inherent alignment with Indic language processing principles enables the seamless integration of labeling functions based on localized dictionaries (e.g., Assamese safety terms, Hindi operational slang) and complex regular expressions. The noisy labels generated by SPEAR are used to fine-tune xlm-roberta-large (550M parameters)⁷.

Hardware Allocation and Docker Orchestration

Running a massive multi-task transformer, an enterprise graph database, and a generative LLM on a single physical node requires exact memory management to prevent catastrophic Out-Of-Memory (OOM) kernel panics. The Docker Compose environment enforces the following memory and compute quotas:

Container / Microservice	Functional Role	VRAM Allocation (Max)	CPU & System RAM Quota	Core Frameworks
prism-layer1-ner	XLM-R Inference for NER and	12 GB (FP16 precision)	4 Cores / 16 GB	PyTorch, HuggingFace,

	Classification			FastAPI
prism-layer3-llm	On-premise Advisory LLM Inference	45 GB (AWQ Quantized)	8 Cores / 32 GB	vLLM, Llama-3-8B / Mistral-7B
prism-layer2-c kg	Symbolic Graph Database & Algorithmic Search	0 GB (CPU only)	8 Cores / 64 GB	Neo4j Graph Data Science
prism-ocr-ingest	On-premise Document Parsing & Layout Analysis	4 GB	4 Cores / 16 GB	Tesseract, PaddleOCR, OpenCV
prism-orchestrator	API Gateway, Message Queue & Caching	0 GB (CPU only)	4 Cores / 8 GB	Nginx, Redis, Celery

(Note: Approximately 19 GB of VRAM is held in reserve to act as a dynamic buffer for managing extreme Key-Value (KV) cache spikes during high-concurrency LLM generation requests).

On-Premise OCR Pipeline

Historical safety records, legacy OISD PDFs, and handwritten shift logs exist largely as scanned, unstructured documents. Tesseract OCR, heavily augmented with custom trained data for Hindi and English scripts, is deployed to extract raw text¹⁹. Post-processing modules utilize computer vision (specifically layout parsing via PaddleOCR) to reconstruct the reading order of multi-column incident reports, filter out bureaucratic boilerplate, and feed sanitized text directly into the Layer 1 inference engine.

4. OIL Safety Causal Knowledge Graph (CKG) Schema

Layer 2 serves as the deterministic, symbolic reasoning engine of PRISM-SIF. The outputs of the Layer 1 NLP model (identified entities and predicted relations) are inherently probabilistic and subject to statistical error. Grounding these extractions into a Causal Knowledge Graph (CKG) ensures that the generative LLM (Layer 3) grounds its safety explanations in factual,

organizationally approved engineering logic, drastically reducing the risk of catastrophic AI hallucination.

The ontology architecture is heavily influenced by the Ontology of Computer-Aided Process Engineering (OntoCAPE), which provides a formal, peer-reviewed semantic framework for modeling chemical engineering domains, plant layouts, and process materials²⁰. The schema is authored in the Web Ontology Language (OWL) and serialized in Turtle (.ttl) format for high human readability and machine parsing efficiency²².

OWL/Turtle Schema Design Specification

The schema is rigidly partitioned into fundamental classes that mirror OISD regulatory taxonomies: Incident, Hazard, Precursor, Equipment, WorkPermit, and Mitigation. The following Turtle specification outlines the core ontological relationships:

Code snippet

```
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix oilsafety: <http://ontology.oilindia.com/safety#> .
```

Core Class Definitions

```
oilsafety:Hazard a owl:Class .
oilsafety:SIFPrecursor a owl:Class ;
    rdfs:subClassOf oilsafety:Hazard ;
    rdfs:comment "A high-energy state or condition with the potential to cause a Serious Injury or Fatality." .
oilsafety:Equipment a owl:Class .
oilsafety:WorkPermit a owl:Class .
oilsafety:HotWorkPermit a owl:Class ;
    rdfs:subClassOf oilsafety:WorkPermit .
oilsafety:ColdWorkPermit a owl:Class ;
    rdfs:subClassOf oilsafety:WorkPermit .
oilsafety:Incident a owl:Class .
oilsafety:Mitigation a owl:Class .
```

Object Properties (Semantic Relations)

```
oilsafety:causes a owl:ObjectProperty ;
    rdfs:domain oilsafety:SIFPrecursor ;
    rdfs:range oilsafety:Incident .
```

```
oilsafety:involvesEquipment a owl:ObjectProperty ;  
  rdfs:domain oilsafety:Incident ;  
  rdfs:range oilsafety:Equipment .
```

```
oilsafety:requiresPermit a owl:ObjectProperty ;  
  rdfs:domain oilsafety:Hazard ;  
  rdfs:range oilsafety:WorkPermit .
```

```
oilsafety:mitigatedBy a owl:ObjectProperty ;  
  rdfs:domain oilsafety:Hazard ;  
  rdfs:range oilsafety:Mitigation .
```

```
# Data Properties (Attributes)  
oilsafety:hasSeverityScore a owl:DatatypeProperty ;  
  rdfs:domain oilsafety:Incident ;  
  rdfs:range xsd:float .
```

```
oilsafety:oisdReference a owl:DatatypeProperty ;  
  rdfs:domain oilsafety:WorkPermit ;  
  rdfs:range xsd:string .
```

Owlready2 Population Pipeline

The population of this graph is dynamic, acting as a real-time digital twin of the organization's hazard state. The Owlready2 Python library is utilized as the bridging mechanism between the Python execution environment (Layer 1 outputs) and the persistent Neo4j backend.

The pipeline executes through a rigid, four-step deterministic process:

1. **Entity Semantic Mapping:** When Layer 1 identifies "Loss of containment in high-pressure separator" as a Hazard, the pipeline executes a semantic similarity search (using cosine similarity of the XLM-R embeddings) against existing oilsafety:Hazard instances in the graph.
2. **Instantiation and Deduplication:** If a match exceeding a 0.92 confidence threshold is found, the instance frequency is updated. If the entity is novel, Owlready2 creates a new instance dynamically, preventing node duplication while allowing the ontology to grow organically.
3. **Relation Extraction:** The co-occurrence of a Hazard and an Incident in a specific syntactical window triggers the creation of an oilsafety:causes relation, weighted by the frequency of co-occurrence across the corpus.
4. **Neo4j Synchronization:** The Owlready2 SQLite backend is periodically synchronized with the Neo4j enterprise graph using the neo4j Python driver. This script programmatically maps OWL Classes to Neo4j Labels and ObjectProperties to Neo4j Relationships, allowing the execution of Cypher queries for advanced pathfinding

algorithms.

5. Core Code Components and Software Architecture

The software architecture is heavily modularized to support continuous learning, low-latency inference, and human-in-the-loop (HITL) oversight.

Joint NER and Classification Architecture

To optimize inference time and minimize VRAM consumption, PRISM-SIF utilizes a multi-task formulation over a single XLM-R base model, preventing the need to run two separate large

transformer models. Let $\mathbf{H} \in \mathbb{R}^{L \times d}$ represent the sequence of contextualized hidden states for an input of length L and dimension d , where $\mathbf{h}_0$ is the representation of the special [CLS] token.

The Sequence Classification Head determines the overall SIF potential of the narrative. The probability of the text containing a SIF precursor is computed via a linear projection of the [CLS] token, followed by a sigmoid activation function to allow for multi-label classification (e.g., classifying a single event as both Environmental and Fatality risks):

$$P(y^{CLS}) = \sigma(\mathbf{W}_{cls}\mathbf{h}_0 + \mathbf{b}_{cls})$$

Simultaneously, the Token Classification Head performs NER. Every token $\mathbf{h}_i$ is passed through a dense layer followed by a softmax activation over the BIO (Begin, Inside, Outside) tagging scheme for the custom safety entity vocabulary:

$$P(y_i^{NER} | \mathbf{h}_i) = \text{softmax}(\mathbf{W}_{ner}\mathbf{h}_i + \mathbf{b}_{ner})$$

The model is optimized using a joint loss function, defined as the weighted sum of the categorical cross-entropy losses of both tasks. The scalar λ allows the system to prioritize entity extraction over general classification during early training phases:

$$\mathcal{L}_{total} = \lambda \mathcal{L}_{NER} + (1 - \lambda) \mathcal{L}_{CLS}$$

Evidence-Based Training (EBT) Verifier Class

The EBTVerifier class acts as the deterministic, programmatic gatekeeper between the probabilistic predictions of Layer 1 and the permanent ingestion of data into Layer 2. It implements strict rule-based logic derived directly from OISD standards.

For instance, if Layer 1 predicts a ConfinedSpaceEntry entity, the EBTVerifier executes a logic block cross-referencing the text for the presence of a corresponding GasTest or BreathingApparatus entity, as mandated by OISD-STD-105¹⁶. If the mitigation entity is absent, the EBTVerifier flags the extraction as an "Unmitigated High-Risk Precursor." Instead of

auto-populating the CKG, this flagged extraction is routed to the HITL pipeline for mandatory human review. This ensures the graph is never polluted with unverified or non-compliant operational logic.

LoRA Incremental Fine-Tuning Pipeline

Safety vernacular and field jargon evolve continuously. Full parameter fine-tuning of the 550M parameter XLM-R model is computationally prohibitive and risks catastrophic forgetting of general language capabilities. PRISM-SIF employs Low-Rank Adaptation (LoRA) for continuous, memory-efficient incremental learning.

By freezing the pre-trained model weights $\mathbf{W}_0 \in \mathbb{R}^{d \times k}$ and injecting trainable rank decomposition matrices $\mathbf{B} \in \mathbb{R}^{d \times r}$ and $\mathbf{A} \in \mathbb{R}^{r \times k}$ (where the rank $r \ll d$), the model is updated efficiently. The forward pass is modified to:

$$\mathbf{h} = \mathbf{W}_0 \mathbf{x} + \Delta \mathbf{W} \mathbf{x} = \mathbf{W}_0 \mathbf{x} + \mathbf{B} \mathbf{A} \mathbf{x}$$

This pipeline runs as an automated weekly Celery task, aggregating verified HITL corrections to generate new, lightweight LoRA adapters that capture the latest operational terminology without requiring a complete model retrain.

HITL React Native Mobile Specifications

To capture field-level feedback and provide the training data necessary for the LoRA pipeline, an offline-first mobile application built on React Native is provided to site supervisors. Due to the intermittent network connectivity prevalent at remote well pads, the application utilizes WatermelonDB for local SQLite caching. The central API pushes flagged SIF precursors (identified by the EBTVerifier) down to the device during periods of connectivity. The supervisor verifies the entity boundaries and hazard classifications via a simplified, swipe-based interface. Upon re-establishing a network connection, the verified JSON payloads are synced back to the central server to feed the active learning loop.

JSON LLM Schemas (Layer 3)

Layer 3 leverages a locally hosted, quantized generative model (e.g., Mistral-7B via vLLM) to generate natural language explanations of the graph data for end-users. Unconstrained LLMs are prone to hallucinating safety advice, which is unacceptable in an industrial context. Therefore, the LLM is forced to output structured JSON using strict grammar constraints.

```

JSON
{
  "title" "SIF Precursor Explanation"
  "type" "object"
  "properties"
    "identified_hazard" "type" "string"
    "causal_chain"
      "type" "array"
      "items" "type" "string"
    "oidsd_violation" "type" "string" "description" "Specific clause from OISD-105 or 117"
    "recommended_action" "type" "string"
    "required" "identified_hazard" "causal_chain" "recommended_action"
}

```

By enforcing this schema, the LLM acts purely as a linguistic rendering engine for the factual data retrieved from the Layer 2 CKG, ensuring that the final advisory presented to the safety officer is highly structured, traceable, and devoid of hallucinations.

6. Evaluation Frameworks

Rigorous, mathematically sound evaluation frameworks must be established for each layer to ensure system integrity prior to organizational deployment.

Layer 1: NLP Evaluation Metrics

Given the extreme class imbalance inherent in safety reporting—where the vast majority of tokens represent background text rather than critical entities—standard accuracy is a highly misleading metric. The evaluation relies strictly on token-level and entity-level performance indicators²³.

Evaluation Metric	Mathematical Definition	Operational Significance in PRISM-SIF
Precision	$TP / (TP +$	Measures how much the model's positive predictions can be trusted. High precision minimizes false alarms, preventing alert fatigue among safety officers ²³ .

Recall	$TP / (TP +$	Measures the ability to find all actual positive samples. In safety applications, maximizing recall is critical to ensure no catastrophic hazard is missed ² .
Micro F1-Score	$2 \times$	The harmonic mean of precision and recall, calculated globally. This serves as the primary optimization metric for the XLM-R multi-task model during backpropagation ²³ .

Furthermore, the generation of confusion matrices is automated during testing to analyze inter-class confusion (e.g., misclassifying an EquipmentFailure entity as an OperatorError), providing targeted insights for refining the labeling functions within the SPEAR weak supervision framework⁹.

Layer 2 and Layer 3 Evaluation Metrics

The CKG is evaluated using standard ontological metrics, including Node/Edge Density (to ensure the graph is sufficiently interconnected) and Logical Consistency²⁰. Logical consistency is continuously evaluated via a semantic reasoner (e.g., Hermit) to ensure there are no contradicting triples in the database.

Evaluating the generative text produced by the Layer 3 LLM requires structured, comparative methodologies:

- **RAG Triad:** Evaluates Context Relevance, Answer Faithfulness, and Answer Relevance against the extracted graph data.
- **Borda Count Method:** Utilized during A/B testing of different LLM architectures. Derived from social choice theory, expert human raters rank the generated explanations. The Borda count treats each evaluation task as a preference voter, aggregating scores to identify the most robust model for safety explanations, a method proven to be highly resilient for comparing NLP systems²⁴.
- **Abstention Rate:** Evaluates the model's ability to refuse answering when the CKG lacks sufficient information. A high abstention rate on out-of-domain queries is a critical safety feature to prevent hallucinations²⁴.

7. Digital Personal Data Protection (DPDP) Act Compliance and AI Governance

The integration of artificial intelligence into critical corporate infrastructure must adhere

rigorously to the Digital Personal Data Protection (DPDP) Act 2023 of India²⁵, as well as emerging international governance standards such as ISO/IEC 42001 (Artificial Intelligence Management System)²⁶.

DPDP Act 2023 Alignment

Field safety reports frequently contain high volumes of personally identifiable information (PII), such as the names of injured workers, shift supervisors, employee ID numbers, and specific duty rosters. While the processing of such data is often legally permitted under the "legitimate purposes" exemption for employment and workplace safety operations, explicit adherence to data minimization principles is mandatory²⁵.

Prior to ingestion by Layer 1, all unstructured text must pass through a highly aggressive redaction pipeline. This layer utilizes deterministic Regular Expressions for identifying alphanumeric IDs and phone numbers, paired with a secondary, lightweight NER model (e.g., a quantized DistilBERT) trained exclusively to identify and redact PERSON and DATE entities. Furthermore, data processed by PRISM-SIF is logically isolated to enforce purpose limitation. The vector embeddings generated by the XLM-R model and the nodes in the CKG are entirely anonymized. The system is mathematically constrained from generating historical trend analyses that identify individual operators as failure points; the analytical focus remains strictly on *process safety deficiencies* and *systemic organizational failures*.

ISO/IEC 42001 AIMS Integration

ISO/IEC 42001 provides a globally recognized, certifiable framework for the responsible, ethical, and transparent governance of AI systems²⁶. PRISM-SIF aligns with these mandates by implementing an Artificial Intelligence Management System (AIMS) that covers the entire software lifecycle.

The standard mandates comprehensive risk management, requiring the systematic identification and mitigation of ethical and security challenges²⁶. PRISM-SIF addresses this through algorithmic transparency; Layer 3 is specifically constrained by the JSON schema to provide causal chains derived directly from the Layer 2 CKG. The system does not act as an opaque "black box"; every safety recommendation is traceable back to a specific OISD standard or historical CSB incident report, satisfying ISO requirements for transparency and accountability²⁸. Furthermore, the LoRA fine-tuning pipeline and the HITL React Native application represent the technical realization of the ISO requirement for ongoing continuous improvement and human oversight²⁷. Attaining full ISO/IEC 42001 certification typically involves a 6 to 10-month audit process requiring rigorous documentation of these exact technical controls³⁰.

8. Technical and Organizational Risk Register & Phase Gate Framework

The deployment of PRISM-SIF carries inherent technical, operational, and organizational risks

that must be cataloged and systematically mitigated within a formalized Risk Register, governing a strict Phase Gate deployment strategy.

Risk Register (RISK-O1 to RISK-O4)

Risk ID	Risk Description	Probability	Operational Impact	Mitigation Strategy
RISK-O1	Code-Switching Degradation: The XLM-R model fails to accurately parse highly contextualized, phonetic Assamese-English vernacular, leading to high false-negative rates on critical SIF precursors.	High	Critical	Implementation of the AI4Bharat SPEAR weak supervision framework to generate highly specific training subsets. Extensive HITL verification loops during Phase 1 deployment.
RISK-O2	Infrastructure OOM Errors: Concurrent API requests to the FastAPI backend, vLLM inference engine, and Neo4j graph overwhelm the 80GB VRAM limit on the single A100 node.	Medium	High	Strict Docker Compose VRAM partitioning. Utilization of AWQ/GPTQ 4-bit quantization for the Layer 3 LLM. Batching inference requests via Celery and Redis message queues.
RISK-O3	CKG Schema Drift: Discrepancies	Medium	Medium	Implementation of an "Unknown

	emerge between the rigid OWL ontology and the fluid nature of newly discovered operational hazards that are not yet codified in OISD standards.			Hazard" sub-graph within Neo4j. Routine semantic alignment reviews by safety engineers to formally instantiate new classes into the OWL schema via Owlready2.
RISK-O4	Lack of User Adoption: Field safety officers reject the system due to interface friction, perceived threats of automation replacing jobs, or a fundamental lack of trust in AI-generated insights.	High	Critical	Deployment of the offline-first React Native app with a frictionless, swipe-based UI. Framing the AI as an "advisory co-pilot" rather than an auditor. Conducting extensive ISO 42001-aligned transparency and ethics training.

Phase Gate Decision Framework

To meticulously manage these risks, project execution is constrained by a Phase Gate model. Progression to subsequent development phases is strictly contingent upon meeting hard, quantifiable engineering exit criteria.

Gate 1: NLP Baseline Feasibility and Weak Supervision Optimization The initial phase focuses on training the Layer 1 XLM-R model using the SPEAR weak supervision framework on

historical CSB, OSHA, and sanitized OIL incident data. Progression through this gate requires the NER head to achieve a Micro F1-score of ≥ 0.88 on a holdout test set of code-switched safety reports. Concurrently, the sequence classification head must achieve a false negative rate of $\leq 2\%$ on identifying SIF potential, guaranteeing that high-risk narratives are not bypassed.

Gate 2: CKG Integration & Logical Consistency Phase 2 involves the deployment of the Owlready2 pipeline mapping Layer 1 outputs to the Neo4j backend, and the ingestion of OISD-105 and OISD-117 logic into the OWL schema. The exit criteria mandate that the HermiT reasoner runs over the populated graph with zero logical inconsistencies. Furthermore, the entity resolution algorithms must achieve 95% accuracy in deduplicating synonymous hazard nodes to prevent graph bloat.

Gate 3: LLM Explainability and Safeguards Phase 3 focuses on the deployment of the on-premise vLLM engine and the rigorous testing of the JSON structured output prompts utilizing the CKG context. Gate clearance requires a Borda count evaluation by three senior safety engineers, ranking the AI explanations as "Highly Actionable" in >85% of test cases. Additionally, the abstention rate must be successfully triggered on 100% of out-of-domain or highly ambiguous queries, proving the hallucination safeguards are functional.

Gate 4: HITL Deployment and Operational System Integration The final pre-production phase involves the rollout of the React Native application to a pilot group of site supervisors and the activation of the LoRA continuous learning pipeline. The system must successfully handle 30 days of continuous operation without experiencing OOM kernel faults. Finally, user analytics must demonstrate that 80% of flagged precursor warnings are actively verified and acted upon by field staff, demonstrating both high system accuracy and requisite user trust. Passing comprehensive DPDP Act data minimization audits is the final requirement for organizational sign-off.

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